

Lab 02

Due Jan 31 5:00 PM PST

AUTHOR

Brandon Cho 301365816

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Instructions

For this homework, provide a rendered R Markdown file in PDF format. You may render the R Markdown file to HTML and then convert the HTML file to PDF using the print function in a web browser. Indicate your student number on the markdown file and make a section for each problem.

Problem 1: My Median

- a. Write an R function called `my_median`. This function should take a single parameter (a vector) and return the median of that vector. Write this function without using the built-in R command `median`. Provide the code for this function.

(4 points)

```
# Example R code
define_my_median <- function(vector) {
  # Code logic here
  n <- length(vector)
  sortedVector <- sort(vector)

  if (n %% 2 == 1) {
    return (sortedVector[(n + 1) / 2])
  }
  else {
    return(mean(sortedVector[n / 2], sortedVector[(n / 2) + 1]))
  }
}
```

- b. Consider the stock prices in the file `AS-N100.tsv` provided in the archive for week 3. These data involve stock prices for 27 symbols (in the column `ticker`). Compute your student number modulo 27 and add one, call this value `a` (i.e., `a` is a number between 1 and 27 inclusive). Imagine a list of the 27 symbols in alphabetical order. You will examine the `a`-th symbol in this list. You may find the symbol using the R code `sort(unique(data$ticker))[a]`. Compute the median opening price of the symbol you examine over all rows in `AS-N100.tsv` involving that symbol, using your `my_median` function. Provide all code and output.

(2 points)

```
## this code doesn't run at the moment, need to modify it

# Example R code to read and process
library(readr)
data <- read_tsv("AS-N100.tsv")
symbols <- sort(unique(data$ticker))
# Compute value of 'a'
# Example of custom median application
a <- 301365816
a <- (a %% 27) + 1

# choose the correct symbol
selected_symbol <- symbols[a]

# Filter data for the selected ticker
filtered_data <- data[data$ticker == selected_symbol, ]

# Compute the median of the opening prices
opening_prices <- filtered_data$open
median_price <- define_my_median(opening_prices)
cat("Selected ticker:", selected_symbol, "\n")
cat("Median opening price:", median_price, "\n")
```

Problem 2: X Marks the Spot

- a. The image [Figure03.png](#) (provided in the archive containing this lab) is a satellite photo of Vancouver and the surrounding area. In this problem, you will mark the location of a library on this image automatically using code.

The upper left-hand corner of this image has approximate GPS coordinates [49.410705, -124.217671](#). The lower right-hand corner of this image has approximate GPS coordinates [47.929083, -121.994887](#). Consider the file [libraries.json](#) provided in the archive for week 3. This file contains information about 21 branches of the Vancouver Public Library.

Write R code to display [Figure03.png](#) and to mark an **x** symbol on the GPS coordinates of the **b**-th library branch. Print the name and address of the library branch directly below the **x**. Here, **b** is your student number modulo 21 plus one (i.e., **b** is a number between 1 and 21 inclusive).

Note that after loading the library [rjson](#) and loading [libraries.json](#) using:

```
libraries <- fromJSON(file = "libraries.json")
```

the coordinates of the **b**-th branch can be accessed with:

```
libraries[["features"]][[b]][["geometry"]]$coordinates
```

The branch name and address can be accessed with:

```
libraries[["features"]][[b]][["properties"]]$maptip
```

Provide your code and the resulting display.

(4 points)

```
# Load required libraries
library(rjson)
library(imager)
```

Loading required package: magrittr

Attaching package: 'imager'

The following object is masked from 'package:magrittr':

add

The following objects are masked from 'package:stats':

convolve, spectrum

The following object is masked from 'package:graphics':

frame

The following object is masked from 'package:base':

save.image

```
# Calculate b
b <- (301365816 %% 21) + 1

# Load libraries data
libraries <- fromJSON(file = "libraries.json")

# Get coords and name
library_coords <- libraries[["features"]][[b]][["geometry"]]$coordinates
library_maptip <- libraries[["features"]][[b]][["properties"]]$maptip

# Read the satellite image
img <- load.image("Figure03.png")

# Set up plot area with geographic bounds
plot(NA,
      xlim = c(-124.217671, -121.994887),
```

```
ylim = c(47.929083, 49.410705),  
axes = FALSE)  
  
# Display the satellite image  
rasterImage(img,  
  xleft = -124.217671,  
  ybottom = 47.929083,  
  xright = -121.994887,  
  ytop = 49.410705)  
  
# Plot X marker at the library's coordinates  
points(x = library_coords[1],  
  y = library_coords[2],  
  pch = 4,  
  col = "red",  
  lwd = 5)  
  
text(library_coords[1],  
  library_coords[2] - 0.1,  
  library_maptip,  
  col = "blue")
```



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